

8 Nonparametric Statistical Methods

8.1 Introduction

Nonparametric statistical methods are statistical procedures that do not rely heavily on strict assumptions about the probability distribution of the population from which samples are obtained.

In many classical statistical methods, it is usually assumed that the data follow a particular probability distribution such as the normal distribution. Methods that rely on such assumptions are called **parametric methods**. Examples include the t -test, ANOVA, and linear regression.

However, in practical applications, these assumptions may not always hold. Data may be skewed, contain outliers, be measured on an ordinal scale, or come from populations with unknown distributions. In such situations, nonparametric methods provide flexible alternatives that remain valid even when distributional assumptions are violated.

Nonparametric methods are therefore designed to make statistical inference with minimal assumptions regarding the underlying population structure.

8.1.1 Alternative Names for Nonparametric Methods

Nonparametric methods are often referred to as:

- **Distribution-free methods**
- **Assumption-light methods**

(a) Distribution-Free Methods

The term **distribution-free** means that the validity of the statistical procedure does not depend on the data following a specific probability distribution such as:

- Normal distribution,
- Exponential distribution,
- Poisson distribution,
- Gamma distribution.

For example, the Mann–Whitney U test can be applied even when the population distribution is unknown or non-normal.

(b) Assumption-Light Methods

The phrase **assumption-light** means that nonparametric procedures require fewer and less restrictive assumptions compared to parametric procedures.

Instead of assuming exact probability models, nonparametric methods often require only basic conditions such as:

- independence of observations,
- random sampling,
- ordinal or continuous measurements.

Because of this flexibility, nonparametric methods are widely used in real-world studies where strict assumptions cannot easily be verified.

8.1.2 Comparison with Parametric Methods

Unlike parametric methods, nonparametric methods do not require assumptions such as:

- normality,
- homogeneity of variance,
- linearity,
- specific functional forms.

Each of these assumptions is explained below.

(a) Normality Assumption

Many parametric methods assume that the population follows a normal distribution.

For example, in the independent sample t -test, it is assumed that:

$$X \sim N(\mu, \sigma^2)$$

where:

- μ is the population mean,
- σ^2 is the population variance.

In practice, however, data may be:

- skewed,

- multimodal,
- heavy-tailed,
- contaminated with outliers.

Nonparametric methods do not require the data to follow a normal distribution and can therefore be applied more broadly.

Example Income data are usually highly skewed to the right. In such cases, the Wilcoxon rank-sum test may be preferred over the two-sample t -test.

(b) Homogeneity of Variance

Parametric procedures such as ANOVA often assume equal variances across groups.

Mathematically, this assumption is expressed as:

$$\sigma_1^2 = \sigma_2^2 = \cdots = \sigma_k^2$$

where:

- σ_i^2 represents the variance of the i -th population.

This assumption is called:

- homoscedasticity,
- equality of variances.

Nonparametric methods generally do not require equal variances, making them more robust when variability differs across groups.

Example Suppose two treatment groups have very different spreads due to unequal variability in patient responses. A Mann–Whitney test may still be appropriate when ANOVA assumptions fail.

(c) Linearity Assumption

In regression analysis, parametric models frequently assume a linear relationship between variables.

For example:

$$Y = \beta_0 + \beta_1 X + \varepsilon$$

This model assumes that the relationship between X and Y is linear.

However, many real-world relationships are nonlinear.

Example The relationship between age and blood pressure may increase rapidly at certain ages and remain stable at others.

Nonparametric regression methods such as kernel smoothing or spline regression can model such relationships without assuming linearity.

(d) Specific Functional Forms

Parametric models often assume that the data-generating process follows a known mathematical structure.

Examples include:

- linear models,
- logistic models,
- exponential growth models.

Nonparametric methods avoid specifying exact functional forms and instead allow the data to determine the structure of the relationship.

This flexibility makes them useful when little prior knowledge exists about the underlying population model.

8.1.3 Motivation for Nonparametric Methods

Nonparametric methods are useful when:

1. The population distribution is unknown.
2. Data are ordinal rather than numerical.
3. Sample sizes are small.
4. Data contain outliers.
5. Assumptions of parametric tests are violated.
6. Data are skewed or heavily tailed.

8.1.4 Why Nonparametric Methods are Important

Nonparametric methods are particularly important because:

1. Real-world data rarely satisfy all parametric assumptions.
2. Small sample sizes make normality assumptions difficult to verify.
3. Many variables are measured on ordinal scales.

4. Outliers can strongly affect parametric estimates.
5. Modern applications such as machine learning often involve highly complex data structures.

Examples of Common Nonparametric Methods

Parametric Method	Nonparametric Alternative
Independent sample t -test	Mann–Whitney U test
Paired t -test	Wilcoxon signed-rank test
One-way ANOVA	Kruskal–Wallis test
Pearson correlation	Spearman rank correlation
Linear regression	Kernel smoothing regression
Polynomial regression	Spline regression

8.1.5 Types of Nonparametric Procedures

- One-Sample Procedures
 - Sign test
 - Wilcoxon signed-rank test
- Two-Sample Procedures
 - Mann–Whitney U test
 - Kolmogorov–Smirnov test
- Multiple-Sample Procedures
 - Kruskal–Wallis test
 - Friedman test
- Association Measures
 - Spearman rank correlation
 - Kendall’s tau

8.2 Rank-Based Tests

Rank-based tests are nonparametric methods that replace raw observations with their ranks and test hypotheses based on these ranks. They are particularly useful when data are not normally distributed or contain outliers.

8.2.1 Mann–Whitney U Test (Wilcoxon Rank-Sum)

Purpose: The Mann–Whitney U test is a nonparametric alternative to the independent samples t-test. It is used to determine whether two independent samples come from the same population or whether one population tends to produce larger observations than the other.

Key Insight (from theory): Rather than comparing means, the test compares **all pairwise observations** across the two samples. Specifically, it evaluates:

$$P(X > Y)$$

that is, the probability that a randomly selected observation from one population exceeds one from the other.

Hypotheses:

- $H_0 : P(X > Y) = \frac{1}{2}$ (distributions are identical)
- $H_1 : P(X > Y) \neq \frac{1}{2}$ (one distribution tends to be larger)

Assumption:

- The two distributions have similar shapes (difference is mainly in location/median)

Test procedure

(a) Rank based approach

- Combine all observations
- Rank them from smallest to largest
- Compare sums of ranks between groups

(b) Pairwise comparison approach

- Compare each x_i with each y_j
- Count number of times $x_i > y_j$
- Total comparisons: $n_x n_y$

Test Statistic

Let:

- U_x = number of times $x_i > y_j$
- U_y = number of times $y_j > x_i$

Then:

$$U = \min(U_x, U_y), \quad \text{where } U_x + U_y = n_x n_y$$

For large samples:

$$\mu_U = \frac{n_x n_y}{2}, \quad \sigma_U = \sqrt{\frac{n_x n_y (n_x + n_y + 1)}{12}}$$

Example 8.2.1

Group A (X): [5, 7, 9]

Group B (Y): [6, 8, 10]

Step 1: Combine and rank

$$5, 6, 7, 8, 9, 10 \Rightarrow \text{Ranks } 1, 2, 3, 4, 5, 6$$

Step 2: Label groups

$$5_X, 6_Y, 7_X, 8_Y, 9_X, 10_Y$$

Step 3: Pairwise comparisons

	6	8	10
5	0	0	0
7	1	0	0
9	1	1	0

Step 4: Compute U statistics

$$U_X = 3, \quad U_Y = 9 - 3 = 6$$

$$U = \min(U_X, U_Y) = 3$$

Step 5: Rank-sum verification

Ranks for Group X: 1, 3, 5

$$R_X = 1 + 3 + 5 = 9$$

$$U_X = R_X - \frac{n_X(n_X + 1)}{2} = 9 - \frac{3(4)}{2} = 9 - 6 = 3$$

Step 6: Exact p-value

Total number of comparisons:

$$n_X n_Y = 3 \times 3 = 9$$

Using Mann–Whitney tables (or enumeration), for $n_X = n_Y = 3$ and $U = 3$:

To compute the exact p-value, we consider all possible ways of assigning ranks under the null hypothesis.

METHOD A Using exact enumeration

Step 6.1: Total possible arrangements

We have 6 observations, so the ranks are:

$$1, 2, 3, 4, 5, 6$$

We choose 3 ranks for Group X (the remaining go to Group Y).

$$\text{Number of ways} = \binom{6}{3} = 20$$

$$n_X = 3, \quad n_Y = 3, \quad N = 6$$

Thus, there are 20 equally likely configurations under H_0 .

Step 6.2: Distribution of the U statistic

For each configuration, compute U . The exact distribution is:

For each selection of ranks for Group X:

$$U_X = R_X - \frac{n_X(n_X + 1)}{2}$$

where:

$$R_X = \text{sum of ranks of Group X}, \quad \frac{n_X(n_X + 1)}{2} = 6$$

Step 3: Enumerate all possibilities

Ranks for X	R_X	$U_X = R_X - 6$
(1,2,3)	6	0
(1,2,4)	7	1
(1,2,5)	8	2
(1,2,6)	9	3
(1,3,4)	8	2
(1,3,5)	9	3
(1,3,6)	10	4
(1,4,5)	10	4
(1,4,6)	11	5
(1,5,6)	12	6
(2,3,4)	9	3
(2,3,5)	10	4
(2,3,6)	11	5
(2,4,5)	11	5
(2,4,6)	12	6
(2,5,6)	13	7
(3,4,5)	12	6
(3,4,6)	13	7
(3,5,6)	14	8
(4,5,6)	15	9

Step 4: Count frequencies of U

Now count how many times each U appears:

U	Frequency
0	1
1	1
2	2
3	3
4	3
5	3
6	3
7	2
8	1
9	1

$$\text{Total} = 20$$

Step 6.3: Observed value

From the data:

$$U_{\text{obs}} = 3$$

Step 6.4: Compute lower-tail probability

$$P(U \leq 3) = \frac{1 + 1 + 2 + 3}{20} = \frac{7}{20} = 0.35$$

Step 6.5: Compute upper-tail probability

By symmetry:

$$P(U \geq 6) = \frac{3 + 2 + 1 + 1}{20} = \frac{7}{20} = 0.35$$

Step 6.6: Two-sided p-value

The two-sided p-value is:

$$p = P(U \leq 3) + P(U \geq 6) = 0.35 + 0.35 = 0.70$$

Final Result:

$$p = 0.70 \quad (\text{two-sided})$$

Interpretation:

Since $p = 0.70 > 0.05$, we fail to reject the null hypothesis. There is no statistically significant evidence of a difference between the two groups.

Step 7: Normal approximation

$$\mu_U = \frac{n_X n_Y}{2} = \frac{9}{2} = 4.5$$

$$\sigma_U = \sqrt{\frac{n_X n_Y (n_X + n_Y + 1)}{12}} = \sqrt{\frac{3 \times 3 \times 7}{12}} = \sqrt{\frac{63}{12}} \approx 2.29$$

$$z = \frac{U - \mu_U}{\sigma_U} = \frac{3 - 4.5}{2.29} \approx -0.66$$

$$p \approx 0.51 \quad (\text{two-sided})$$

Interpretation:

Although all observations in Group B are larger than those in Group A, the sample size is very small. The p-value is large ($p > 0.05$), so we fail to reject the null hypothesis.

Conclusion:

There is no statistically significant evidence of a difference between the two groups. However, the consistent ordering suggests a practical difference, and a larger sample would likely detect significance.

METHOD B: Using Mann–Whitney tables**Step 1: Identify sample sizes**

$$n_X = 3, \quad n_Y = 3$$

Step 2: Observed test statistic

From earlier computation:

$$U_{\text{obs}} = 3$$

Step 3: Use Mann–Whitney U critical value table

From standard Mann–Whitney tables (exact distribution):

- For $n_X = 3, n_Y = 3$
- The possible values of U range from 0 to 9

The table provides cumulative probabilities $P(U \leq u)$.

At significance level $\alpha = 0.05$ (two-sided), the critical value from tables is:

$$U_{\text{critical}} = 0$$

Decision rule:

- Reject H_0 if $U \leq 0$

Since:

$$U_{\text{obs}} = 3 > 0$$

we fail to reject H_0 .

Interpretation

The table-based approach confirms the exact enumeration result. The observed value $U = 3$ is not extreme enough to indicate a statistically significant difference between the two groups.

Key Teaching Insight

Mann–Whitney tables already contain the enumeration results. They save computation by directly giving $P(U \leq u)$ for given (n_X, n_Y) .

METHOD C Using Normal Approximation**Step 1: Given data**

$$n_X = 3, \quad n_Y = 3, \quad U = 3$$

Step 2: Compute mean and standard deviation

$$\mu_U = \frac{n_X n_Y}{2} = \frac{3 \times 3}{2} = 4.5$$

$$\sigma_U = \sqrt{\frac{n_X n_Y (n_X + n_Y + 1)}{12}} = \sqrt{\frac{3 \times 3 \times 7}{12}} = \sqrt{\frac{63}{12}} \approx 2.29$$

Step 3: Apply continuity correction

Since U is discrete, we use:

$$U_c = U + 0.5 = 3 + 0.5 = 3.5$$

Step 4: Compute z-statistic

$$z = \frac{U_c - \mu_U}{\sigma_U} = \frac{3.5 - 4.5}{2.29} = \frac{-1}{2.29} \approx -0.44$$

Step 5: Compute p-value

From standard normal tables:

$$P(Z \leq -0.44) \approx 0.33$$

Two-sided p-value:

$$p = 2 \times 0.33 = 0.66$$

Final Result:

$p \approx 0.66 \quad (\text{two-sided})$

Interpretation:

Since $p = 0.66 > 0.05$, we fail to reject the null hypothesis. There is no statistically significant evidence of a difference between the two groups.

Important Note:

- The normal approximation is less accurate for very small samples
- The exact p-value (≈ 0.70) is more reliable in this case

Mann Whitney U Test - Critical Values Table

n ₂	Probability Range	n ₁																	
		3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
3	0.05	0	0	0	1	1	2	2	3	3	4	4	5	5	6	6	7	7	8
	0.01	0	0	0	0	0	0	0	0	0	1	1	1	2	2	2	2	3	3
4	0.05	0	0	1	2	3	4	4	5	6	7	8	9	10	11	11	12	13	14
	0.01	0	0	0	0	0	1	1	2	2	3	3	4	5	5	6	6	7	8
5	0.05	0	1	2	3	5	6	7	8	9	11	12	13	14	15	17	18	19	20
	0.01	0	0	0	1	1	2	3	4	5	6	7	7	8	9	10	11	12	13
6	0.05	1	2	3	5	6	8	10	11	13	14	16	17	19	21	22	24	25	27
	0.01	0	0	1	2	3	4	5	6	7	9	10	11	12	13	15	16	17	18
7	0.05	1	3	5	6	8	10	12	14	16	18	20	22	24	26	28	30	32	34
	0.01	0	0	1	3	4	6	7	9	10	12	13	15	16	18	19	21	22	24
8	0.05	2	4	6	8	10	13	15	17	19	22	24	26	29	31	34	36	38	41
	0.01	0	1	2	4	6	7	9	11	13	15	17	18	20	22	24	26	28	30
9	0.05	2	4	7	10	12	15	17	20	23	26	28	31	34	37	39	42	45	48
	0.01	0	1	3	5	7	9	11	13	16	18	20	22	24	27	29	31	33	36
10	0.05	3	5	8	11	14	17	20	23	26	29	33	36	39	42	45	48	52	55
	0.01	0	2	4	6	9	11	13	16	18	21	24	26	29	31	34	37	39	42
11	0.05	3	6	9	13	16	19	23	26	30	33	37	40	44	47	51	55	58	62
	0.01	0	2	5	7	10	13	16	18	21	24	27	30	33	36	39	42	45	48
12	0.05	4	7	11	14	18	22	26	29	33	37	41	45	49	53	57	61	65	69
	0.01	1	3	6	9	12	15	18	21	24	27	31	34	37	41	44	47	51	54
13	0.05	4	8	12	16	20	24	28	33	37	41	45	50	54	59	63	67	72	76
	0.01	1	3	7	10	13	17	20	24	27	31	34	38	42	45	49	53	56	60
14	0.05	5	9	13	17	22	26	31	36	40	45	50	55	59	64	67	74	78	83
	0.01	1	4	7	11	15	18	22	26	30	34	38	42	46	50	54	58	63	67
15	0.05	5	10	14	19	24	29	34	39	44	49	54	59	64	70	75	80	85	90
	0.01	2	5	8	12	16	20	24	29	33	37	42	46	51	55	60	64	69	73
16	0.05	6	11	15	21	26	31	37	42	47	53	59	64	70	75	81	86	92	98
	0.01	2	5	9	13	18	22	27	31	36	41	45	50	55	60	65	70	74	79
17	0.05	6	11	17	22	28	34	39	45	51	57	63	67	75	81	87	93	99	105
	0.01	2	6	10	15	19	24	29	34	39	44	49	54	60	65	70	75	81	86
18	0.05	7	12	18	24	30	36	42	48	55	61	67	74	80	86	93	99	106	112
	0.01	2	6	11	16	21	26	31	37	42	47	53	58	64	70	75	81	87	92
19	0.05	7	13	19	25	32	38	45	52	58	65	72	78	85	92	99	106	113	119
	0.01	3	7	12	17	22	28	33	39	45	51	56	63	69	74	81	87	93	99
20	0.05	8	14	20	27	34	41	48	55	62	69	76	83	90	98	105	112	119	127
	0.01	3	8	13	18	24	30	36	42	48	54	60	67	73	79	86	92	99	105

Example 8.2.2

The following data shows the age at diagnosis of type II diabetes in young adults. Is the age at diagnosis different for males and females?

- Males: 19 22 16 29 24
- Females: 20 11 17 12

Solution

1. Arrange in order of magnitude:

Age	11	12	16	17	19	20	22	24	29
M/F	F	F	M	F	M	F	M	M	M
$M > F$			2		3		4	4	4
$F > M$	0	0		1		2			

2. Affix M or F to each observation (see above).
3. Under each M write the number of Fs to the left of it; under each F write the number of Ms to the left of it (see above).
4. Compute:

$$U_M = 2 + 3 + 4 + 4 + 4 = 17, \quad U_F = 0 + 0 + 1 + 2 = 3$$

5. Calculate:

$$U = \min(U_M, U_F) = 3$$

6. Using tables for the Mann–Whitney U test, we obtain a two-sided p-value:

$$p = 0.11$$

7. Using the normal approximation:

$$z = \frac{U - \frac{n_x n_y}{2}}{\sqrt{\frac{n_x n_y (n_x + n_y + 1)}{12}}} = \frac{3 - 10}{\sqrt{50/3}} = -1.715$$

This gives a two-sided p-value of:

$$p = 0.09$$

Interpretation

The exact test and the normal approximation give similar results. We conclude that there is no strong evidence that the age at diagnosis differs between males and females. However, the results are borderline, and the lack of statistical significance may be due to the small sample size.

The median age at diagnosis is 14.5 years for females and 22 years for males, which suggests a substantial practical difference. A larger study would be advisable to confirm this finding.

Example 8.2.3**Mann–Whitney U Test with Ties****Data**

Group A (X): 5, 7, 7, 10

Group B (Y): 6, 7, 8, 9

Step 1: Combine and rank (handle ties)

5, 6, 7, 7, 7, 8, 9, 10

Ranks without ties would be:

1, 2, 3, 4, 5, 6, 7, 8

But there are three tied values at 7, so assign average rank:

$$\text{Ranks for 7s} = \frac{3 + 4 + 5}{3} = 4$$

Final ranked data:

Value	5	6	7	7	7	8	9	10
Group	X	Y	X	X	Y	Y	Y	X
Rank	1	2	4	4	4	6	7	8

Step 2: Compute rank sums

Group X ranks: 1, 4, 4, 8

$$R_X = 1 + 4 + 4 + 8 = 17$$

Step 3: Compute U statistic

$$n_X = 4, \quad n_Y = 4$$

$$U_X = R_X - \frac{n_X(n_X + 1)}{2} = 17 - \frac{4 \times 5}{2} = 17 - 10 = 7$$

$$U_Y = n_X n_Y - U_X = 16 - 7 = 9$$

$$U = \min(7, 9) = 7$$

Step 4: Normal approximation with tie correction

$$\mu_U = \frac{n_X n_Y}{2} = \frac{16}{2} = 8$$

Tie correctionThere is one tie group of size $t = 3$

$$\sum(t^3 - t) = 3^3 - 3 = 27 - 3 = 24$$

Total observations:

$$N = 8$$

$$\begin{aligned}\sigma_U &= \sqrt{\frac{n_X n_Y}{N(N-1)} \left[\frac{N^3 - N}{12} - \frac{\sum(t^3 - t)}{12} \right]} \\ &= \sqrt{\frac{16}{8 \times 7} \left[\frac{512 - 8}{12} - \frac{24}{12} \right]} \\ &= \sqrt{\frac{16}{56} \left[\frac{504}{12} - 2 \right]} \\ &= \sqrt{\frac{16}{56}(42 - 2)} = \sqrt{\frac{16}{56} \times 40} = \sqrt{11.43} \approx 3.38\end{aligned}$$

Step 5: Compute z-statistic (with continuity correction)

$$z = \frac{U + 0.5 - \mu_U}{\sigma_U} = \frac{7.5 - 8}{3.38} \approx -0.15$$

Step 6: Compute p-value

$$P(Z \leq -0.15) \approx 0.44$$

$$p = 2 \times 0.44 = 0.88$$

Final Result:

$$p \approx 0.88$$

Interpretation

There is no statistically significant difference between the two groups. The presence of ties slightly reduces the variability of ranks, which is accounted for in the adjusted variance.

Key Takeaways

- Ties are handled by assigning average ranks
- Variance must be adjusted when using normal approximation
- Ignoring tie correction can lead to incorrect inference

8.2.2 Wilcoxon Signed-Rank Test (Paired Data)

Purpose: The Wilcoxon Signed-Rank Test is a nonparametric alternative to the paired t-test. It is used to compare two related samples, such as:

- Before-and-after measurements
- Matched or paired observations
- Repeated measurements on the same subjects

Unlike the paired t-test, it does not assume normality of differences. Instead, it uses the ranks of the absolute differences.

Hypotheses:

- H_0 : The median of the differences is zero
- H_1 : The median of the differences is not zero

Test Procedure

1. Compute paired differences:

$$d_i = \text{After}_i - \text{Before}_i$$

2. Remove zero differences (if any)
3. Take absolute values $|d_i|$
4. Rank the absolute differences
5. Assign signs to ranks based on original differences
6. Compute:

$$W^+ = \text{sum of positive ranks}, \quad W^- = \text{sum of negative ranks}$$

7. Test statistic:

$$W = \min(W^+, W^-)$$

Example 8.2.4

Before: [10, 11, 9, 12, 7, 13]

After: [13, 14, 15, 13, 12, 10]

Step 1: Compute paired differences

$$d_i = \text{After} - \text{Before}$$

$$d = [3, 3, 6, 1, 5, -3]$$

Step 2: Compute absolute differences

$$|d| = [3, 3, 6, 1, 5, 3]$$

Step 3: Rank the absolute differences

The absolute differences in ascending order are:

$$1, 3, 3, 3, 5, 6$$

Assign ranks:

- $1 \rightarrow 1$
- The three tied values 3, 3, 3 occupy ranks 2, 3, 4, hence each receives:

$$\frac{2 + 3 + 4}{3} = 3$$

- $5 \rightarrow 5$
- $6 \rightarrow 6$

d_i	$ d_i $	Rank	Signed Rank
3	3	3	+3
3	3	3	+3
6	6	6	+6
1	1	1	+1
5	5	5	+5
-3	3	3	-3

Step 4: Compute the test statistic

Sum of positive ranks:

$$W^+ = 3 + 3 + 6 + 1 + 5 = 18$$

Sum of negative ranks:

$$W^- = 3$$

The Wilcoxon Signed-Rank test statistic is:

$$W = \min(W^+, W^-)$$

$$W = \min(18, 3) = 3$$

Interpretation:

Most of the differences are positive, indicating that the values after treatment tend to be larger than the values before treatment. The small value of the test statistic suggests evidence of a treatment effect.

Example 8.2.5
Example 2 (10 Paired Observations)

Before: [15, 18, 20, 22, 19, 17, 21, 23, 16, 24]

After: [17, 16, 22, 21, 20, 15, 23, 22, 18, 23]

Step 1: Compute differences

$$d = [2, -2, 2, -1, 1, -2, 2, -1, 2, -1]$$

Step 2: Absolute differences

$$|d| = [2, 2, 2, 1, 1, 2, 2, 1, 2, 1]$$

Step 3: Rank absolute differences

Ranks are assigned based on $|d|$, with ties averaged.

d_i	$ d_i $	Rank	Signed Rank
2	2	7	+7
-2	2	7	-7
2	2	7	+7
-1	1	2.5	-2.5
1	1	2.5	+2.5
-2	2	7	-7
2	2	7	+7
-1	1	2.5	-2.5
2	2	7	+7
-1	1	2.5	-2.5

Explanation of ranks:

- Four values of $|d| = 1 \Rightarrow$ ranks 1, 2, 3, 4 $\Rightarrow$ average = 2.5
- Six values of $|d| = 2 \Rightarrow$ ranks 5, 6, 7, 8, 9, 10 $\Rightarrow$ average = 7

Step 4: Compute test statistics

$$W^+ = 7 + 7 + 2.5 + 7 + 7 = 30.5$$

$$W^- = 7 + 2.5 + 7 + 2.5 + 2.5 = 21.5$$

$$W = \min(W^+, W^-) = 21.5$$

Interpretation: Since both positive and negative ranks are present, there is no clear directional dominance. The result suggests a mixed effect, and further comparison with critical values or p -value is required to determine statistical significance.

Step 5: Decision Rule and Statistical Significance

Sample size: $n = 10$

Test statistic:

$$W = \min(W^+, W^-) = 21.5$$

(a) Critical Value Approach

From Wilcoxon signed-rank tables at $\alpha = 0.05$ (two-tailed) and $n = 10$:

$$W_{\text{critical}} = 8$$

Decision rule:

$$\text{Reject } H_0 \text{ if } W \leq 8.$$

Since:

$$21.5 > 8,$$

we **fail to reject** H_0 .

(b) Normal Approximation (for larger samples)

$$\mu_W = \frac{n(n+1)}{4} = \frac{10 \cdot 11}{4} = 27.5$$

$$\sigma_W = \sqrt{\frac{n(n+1)(2n+1)}{24}} = \sqrt{\frac{10 \cdot 11 \cdot 21}{24}} = \sqrt{96.25} \approx 9.81$$

Standardize:

$$Z = \frac{W - \mu_W}{\sigma_W} = \frac{21.5 - 27.5}{9.81} = \frac{-6}{9.81} \approx -0.61$$

From standard normal tables:

$$p \approx 0.54$$

Final Conclusion:

Since the p -value is large ($p > 0.05$), we **fail to reject the null hypothesis**.

Interpretation: There is no statistically significant difference between the before and after measurements. The observed differences are likely due to random variation rather than a systematic effect.

Wilcoxon Signed-Rank Test Critical Values Table

Reject H_0 if the test value is less than or equal to the value given in the table.

n	One tailed, $\alpha = 0.05$	$\alpha = 0.025$	$\alpha = 0.01$	$\alpha = 0.005$
	Two tailed, $\alpha = 0.10$	$\alpha = 0.05$	$\alpha = 0.02$	$\alpha = 0.01$
5	1	--	--	--
6	2	1	--	--
7	4	2	0	--
8	6	4	2	1
9	8	6	3	2
10	11	8	5	3
11	14	11	7	5
12	17	14	10	7
13	21	17	13	10
14	26	21	16	13
15	30	25	20	16
16	36	30	24	19
17	41	35	28	23
18	47	40	33	28
19	54	46	38	32
20	60	52	43	37
21	68	59	49	43
22	75	66	56	49
23	83	73	62	55
24	92	81	69	61
25	101	90	77	68
26	110	98	85	76
27	120	107	93	84
28	130	117	102	92
29	141	127	111	100
30	152	137	120	109



Example 2 (Larger Dataset)

Suppose we have blood pressure readings before and after treatment for 20 patients.

- Compute differences
- Rank absolute differences
- Apply normal approximation if $n > 10$

For large samples:

$$\mu_W = \frac{n(n+1)}{4}, \quad \sigma_W = \sqrt{\frac{n(n+1)(2n+1)}{24}}$$

Practical: Mann–Whitney U Test in R

Illustration 8.2.1

Background of the Dataset

The dataset `PlantGrowth` is available in base R. It contains measurements of plant yield under different treatment conditions.

- **weight**: Dry weight of plants
- **group**: Treatment group (`ctrl`, `trt1`, `trt2`)

This dataset is suitable for demonstrating nonparametric tests because:

- It contains small samples
- It may include tied values
- It involves comparison of independent groups

Objective

To test whether there is a difference in plant growth between treatment groups using the Mann–Whitney U test.

Step 1: Load and Explore Data

```
# Load dataset
data(PlantGrowth)

# View structure
str(PlantGrowth)

# Preview first rows
```

```
head(PlantGrowth)
```

```
# Summary statistics by group
aggregate(weight ~ group, data = PlantGrowth, summary)
```

Step 2: Visual Exploration

```
boxplot(weight ~ group, data = PlantGrowth,
        main = "Plant Growth by Treatment Group",
        xlab = "Group", ylab = "Weight",
        col = c("lightblue", "lightgreen", "lightpink"))
```

Interpretation:

- Compare medians across groups
- Observe spread and overlap of distributions

Step 3: Extract Two Groups

```
# Compare control vs treatment 1
x <- PlantGrowth$weight[PlantGrowth$group == "ctrl"]
y <- PlantGrowth$weight[PlantGrowth$group == "trt1"]
```

Step 4: Perform Mann–Whitney U Test

```
wilcox.test(x, y)
```

Step 5: Exact Test

```
wilcox.test(x, y, exact = TRUE)
```

Step 6: Check for Ties

```
any(duplicated(PlantGrowth$weight))
```

Step 7: Normal Approximation

```
wilcox.test(x, y, exact = FALSE)
```

Typical Output

Wilcoxon rank sum test with continuity correction

W = 58, p-value = 0.06

Interpretation of Results

- The p-value is approximately 0.06
- At significance level $\alpha = 0.05$, we fail to reject H_0
- There is weak evidence of a difference in plant growth between the groups

Student Tasks

Task 1: Hypothesis Testing

- Compare `ctrl` vs `trt1`
- Compare `ctrl` vs `trt2`
- Report test statistic and p-value
- State your conclusion

Task 2: Visualization

- Create boxplots for all groups
- Identify which group appears to have higher values

Task 3: Investigate Ties

- Check if tied values exist
- Explain how ties affect ranking

Task 4: Exact vs Approximate

- Run both:

```
wilcox.test(x, y, exact = TRUE)  
wilcox.test(x, y, exact = FALSE)
```
- Compare the results

Task 5: Interpretation

- What does the test measure?
- Are the groups significantly different?
- Is the difference practically important?

Extension: Multiple Groups

Kruskal–Wallis Test

```
kruskal.test(weight ~ group, data = PlantGrowth)
```

Post-hoc Pairwise Comparisons

```
pairwise.wilcox.test(PlantGrowth$weight, PlantGrowth$group)
```

8.2.3 Kruskal–Wallis Test (k-Sample)

Purpose: The Kruskal–Wallis test is a nonparametric alternative to one-way ANOVA. It is used to compare more than two independent groups when the assumptions of ANOVA (especially normality) are not satisfied.

It tests whether the distributions of the groups are identical or whether at least one group tends to produce larger (or smaller) observations.

Hypotheses:

- H_0 : All groups have identical distributions (same median)
- H_1 : At least one group differs

Test Idea

- Combine all observations from all groups
- Rank the data from smallest to largest
- Compute average ranks for each group
- Compare these average ranks

If group medians differ, their average ranks will differ significantly.

Test Statistic

Let:

- k = number of groups
- n_i = sample size of group i
- $N = \sum n_i$ = total sample size
- R_i = sum of ranks for group i

The Kruskal–Wallis statistic is:

$$H = \frac{12}{N(N+1)} \sum_{i=1}^k \frac{R_i^2}{n_i} - 3(N+1)$$

With ties: apply a correction factor.

For large samples:

$$H \sim \chi_{k-1}^2$$

Example 1 (Small Dataset: Full Computation)

Group A: [5, 6, 7]

Group B: [8, 9, 10]

Group C: [11, 12, 13]

Step 1: Combine and rank

5, 6, 7, 8, 9, 10, 11, 12, 13 $\Rightarrow$ Ranks 1, 2, 3, 4, 5, 6, 7, 8, 9

Step 2: Compute rank sums

- Group A: $1 + 2 + 3 = 6$
- Group B: $4 + 5 + 6 = 15$
- Group C: $7 + 8 + 9 = 24$

$$n_1 = n_2 = n_3 = 3, \quad N = 9$$

Step 3: Compute test statistic

$$\begin{aligned} H &= \frac{12}{9(10)} \left(\frac{6^2}{3} + \frac{15^2}{3} + \frac{24^2}{3} \right) - 3(10) \\ &= \frac{12}{90} (12 + 75 + 192) - 30 \\ &= \frac{12}{90} (279) - 30 = 37.2 - 30 = 7.2 \end{aligned}$$

Step 4: Decision

Degrees of freedom:

$$df = k - 1 = 2$$

Critical value at $\alpha = 0.05$:

$$\chi_{0.05,2}^2 \approx 5.99$$

$$7.2 > 5.99 \Rightarrow \text{Reject } H_0$$

Interpretation: There is strong evidence that at least one group differs.

Upper Critical Values for the Kruskal-Wallis Test

(k samples)

- Notes: 1. In the table below, the critical values give significance levels as close as possible to, but not exceeding the nominal α . The actual levels of significance are in brackets.
2. When the table below is not applicable and $k = 3$ with the three group sizes above 5 or $k > 3$ with all group sizes above 4, use the following approximation:

$$h_{\alpha} = \chi_{\alpha}^2, \quad \text{df} = k - 1.$$

Group Sizes			Nominal size α			
			0.10	0.05	0.025	0.01
2	2	2	4.571 (.06667)	--	--	--
3	2	1	4.286 (.10000)	--	--	--
3	2	2	4.500 (.06667)	4.714 (.04762)	--	--
3	3	1	4.571 (.10000)	5.143 (.04286)	--	--
3	3	2	4.556 (.10000)	5.361 (.03214)	5.556 (.02500)	--
3	3	3	4.622 (.10000)	5.600 (.05000)	5.956 (.02500)	7.200 (.00357)
4	2	1	4.500 (.07619)	--	--	--
4	2	2	4.458 (.10000)	5.333 (.03333)	5.500 (.02381)	--
4	3	1	4.056 (.09286)	5.208 (.05000)	5.833 (.02143)	--
4	3	2	4.511 (.09841)	5.444 (.04603)	6.000 (.02381)	6.444 (.00794)
4	3	3	4.709 (.09238)	5.791 (.04571)	6.155 (.02476)	6.745 (.01000)
4	4	1	4.167 (.08254)	4.967 (.04762)	6.167 (.02222)	6.667 (.00952)
4	4	2	4.555 (.09778)	5.455 (.04571)	6.327 (.02413)	7.036 (.00571)
4	4	3	4.545 (.09905)	5.598 (.04866)	6.394 (.02476)	7.144 (.00970)
4	4	4	4.654 (.09662)	5.692 (.04866)	6.615 (.02424)	7.654 (.00762)
5	2	1	4.200 (.09524)	5.000 (.04762)	--	--
5	2	2	4.373 (.08995)	5.160 (.03439)	6.000 (.01852)	6.533 (.00794)
5	3	1	4.018 (.09524)	4.960 (.04762)	6.044 (.01984)	--
5	3	2	4.651 (.09127)	5.251 (.04921)	6.004 (.02460)	6.909 (.00873)
5	3	3	4.533 (.09697)	5.648 (.04892)	6.315 (.02121)	7.079 (.00866)
5	4	1	3.987 (.09841)	4.985 (.04444)	5.858 (.02381)	6.955 (.00794)
5	4	2	4.541 (.09841)	5.273 (.04877)	6.068 (.02482)	7.205 (.00895)
5	4	3	4.549 (.09892)	5.656 (.04863)	6.410 (.02496)	7.445 (.00974)
5	4	4	4.668 (.09817)	5.657 (.04906)	6.673 (.02429)	7.760 (.00946)
5	5	1	4.109 (.08586)	5.127 (.04618)	6.000 (.02165)	7.309 (.00938)
5	5	2	4.623 (.09704)	5.338 (.04726)	6.346 (.02489)	7.338 (.00962)
5	5	3	4.545 (.09965)	5.705 (.04612)	6.549 (.02436)	7.578 (.00968)
5	5	4	4.523 (.09935)	5.666 (.04931)	6.760 (.02490)	7.823 (.00978)
5	5	5	4.560 (.09952)	5.780 (.04878)	6.740 (.02475)	8.000 (.00946)

Group Sizes			Nominal size α			
			0.10	0.05	0.025	0.01
6	1	1	- -	- -	- -	- -
6	2	1	4.200 (.09524)	4.822 (.04762)	5.600 (.02381)	- -
6	2	2	4.545 (.08889)	5.345 (.03810)	5.745 (.02063)	6.655 (.00794)
6	3	1	3.909 (.09524)	4.855 (.05000)	5.945 (.02143)	6.873 (.00714)
6	3	2	4.682 (.08528)	5.348 (.04632)	6.136 (.02294)	6.970 (.00909)
6	3	3	4.590 (.09773)	5.615 (.04968)	6.436 (.02229)	7.410 (.00779)
6	4	1	4.038 (.09437)	4.947 (.04675)	5.856 (.02424)	7.106 (.00866)
6	4	2	4.494 (.09986)	5.340 (.04906)	6.186 (.02453)	7.340 (.00967)
6	4	3	4.604 (.09997)	5.610 (.04862)	6.538 (.02498)	7.500 (.00966)
6	4	4	4.595 (.09847)	5.681 (.04881)	6.667 (.02495)	7.795 (.00990)
6	5	1	4.128 (.09271)	4.990 (.04726)	5.951 (.02453)	7.182 (.00974)
6	5	2	4.596 (.09807)	5.338 (.04729)	6.196 (.02481)	7.376 (.00982)
6	5	3	4.535 (.09932)	5.602 (.04956)	6.667 (.02452)	7.590 (.00999)
6	5	4	4.522 (.09974)	5.661 (.04991)	6.750 (.02473)	7.936 (.00998)
6	5	5	4.547 (.09835)	5.729 (.04973)	6.788 (.02484)	8.028 (.00988)
6	6	1	4.000 (.09774)	4.945 (.04779)	5.923 (.02381)	7.121 (.00932)
6	6	2	4.438 (.09824)	5.410 (.04993)	6.210 (.02443)	7.467 (.00982)
6	6	3	4.558 (.09948)	5.625 (.04999)	6.725 (.02462)	7.725 (.00985)
6	6	4	4.548 (.09982)	5.724 (.04950)	6.812 (.02458)	8.000 (.00998)
6	6	5	4.542 (.09987)	5.765 (.04993)	6.848 (.02489)	8.124 (.00990)
6	6	6	4.643 (.09874)	5.801 (.04905)	6.889 (.02493)	8.222 (.00994)
7	7	7	4.594 (.09933)	5.819 (.04911)	6.954 (.02446)	8.378 (.00992)
8	8	8	4.595 (.09933)	5.805 (.04973)	6.995 (.02485)	8.465 (.00991)

Example 2 (Larger Dataset)

Suppose we compare incomes across three regions with 30 observations each.

- Rank all 90 observations
- Compute rank sums
- Use chi-square approximation

Large sample sizes make the approximation more accurate.

Practical Example in R

Dataset: `iris`

Background: The dataset contains measurements of iris flowers from three species:

- `setosa`
- `versicolor`
- `virginica`

We test whether sepal length differs across species.

```
data(iris)
```

```
kruskal.test(Sepal.Length ~ Species, data = iris)
```

Typical Output:

```
Kruskal-Wallis rank sum test
```

```
Kruskal-Wallis chi-squared = 96.9, df = 2, p-value < 2.2e-16
```

Interpretation:

- Very small p-value (< 0.05)
- Strong evidence that at least one species differs
- Differences in sepal length across species are significant

Post-hoc Analysis

The Kruskal–Wallis test does not indicate which groups differ. Perform pairwise comparisons:

```
pairwise.wilcox.test(iris$Sepal.Length, iris$Species)
```

Interpretation:

- Identifies which specific pairs differ
- Adjusts for multiple comparisons

Visualization

```
boxplot(Sepal.Length ~ Species, data = iris,  
        main = "Sepal Length by Species",  
        col = c("lightblue", "lightgreen", "lightpink"))
```

Interpretation:

- Compare medians visually
- Check spread and overlap

Student Tasks

1. Apply Test

- Test differences in Petal.Length across species

2. Compare Methods

- Perform ANOVA:

```
aov_model <- aov(Sepal.Length ~ Species, data = iris)  
summary(aov_model)
```
- Compare with Kruskal–Wallis results

3. Visualization

- Create boxplots
- Interpret differences visually

4. Interpretation

- Explain results in plain language
- Identify which species differ

Key Insights

- Nonparametric alternative to ANOVA
- Based on ranks, not raw data
- Robust to non-normality and outliers
- Requires post-hoc tests to locate differences

8.3 Goodness-of-Fit Tests

A goodness-of-fit test determines whether a sample of observations is consistent with a specified probability distribution.

The null hypothesis usually states that the data come from a particular distribution.

General Hypotheses Let $F_0(x)$ denote a specified distribution function.

We test:

$$H_0 : F(x) = F_0(x)$$

against

$$H_1 : F(x) \neq F_0(x)$$

8.3.1 Chi-Square Goodness-of-Fit Test

The Chi-Square Goodness-of-Fit Test is one of the most widely used nonparametric statistical procedures for determining whether a set of observed data follows a specified theoretical probability distribution.

The test was developed by Karl Pearson and is particularly useful when data are classified into categories or groups. It compares the frequencies actually observed in the sample with the frequencies that would be expected if the hypothesized distribution were true.

The test is called a “goodness-of-fit” test because it measures how well a theoretical distribution fits the observed data.

Purpose of the Test

The Chi-Square Goodness-of-Fit Test is used to determine whether sample data are consistent with a hypothesized population distribution.

The test may be used to assess whether data follow:

- Binomial distribution

- Poisson distribution
- Normal distribution
- Uniform distribution
- Exponential distribution
- Any other specified probability distribution

General Hypotheses

Suppose that the population distribution is hypothesized to follow a specified distribution $F_0(x)$.

The hypotheses are:

H_0 : The data follow the specified distribution

against

H_1 : The data do not follow the specified distribution

Idea of the Test

The main idea behind the Chi-Square Goodness-of-Fit Test is to compare:

- the observed frequencies obtained from the sample, and
- the expected frequencies predicted by the hypothesized distribution.

If the observed frequencies are close to the expected frequencies, then the discrepancy is small and the null hypothesis is likely to be accepted.

However, if the differences are large, then the observed data may not have come from the hypothesized distribution, leading to rejection of the null hypothesis.

Thus, the test measures the extent of disagreement between observed and expected frequencies.

Observed Frequencies

The observed frequency refers to the actual number of observations falling into a particular category or class interval.

Suppose the data are divided into k classes.

Then:

O_i = Observed frequency in class i

Examples include:

- Number of students obtaining grade A
- Number of defective items in a batch
- Number of accidents occurring per day
- Number of patients in each age group

Expected Frequencies

The expected frequency is the number of observations expected in each category if the null hypothesis is true.

Suppose:

- n = total sample size
- p_i = theoretical probability of class i

Then the expected frequency for class i is:

$$E_i = np_i$$

The expected frequencies are therefore computed from the assumed probability distribution.

Interpretation of Differences

The quantity:

$$O_i - E_i$$

measures the deviation between the observed and expected frequencies.

- Small deviations suggest good agreement with the hypothesized distribution.
- Large deviations suggest poor agreement.

However, deviations alone are insufficient because larger expected frequencies naturally tend to produce larger differences.

To standardize these differences, the deviations are divided by the expected frequencies.

Test Statistic

The Chi-Square test statistic is given by:

$$\chi^2 = \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i}$$

where:

- O_i = observed frequency in class i
- E_i = expected frequency in class i
- k = number of categories or classes

Explanation of the Formula

The formula works as follows:

1. Compute the difference between observed and expected frequencies:

$$O_i - E_i$$

2. Square the difference:

$$(O_i - E_i)^2$$

Squaring ensures:

- all contributions are positive,
- large discrepancies receive more weight.

3. Divide by the expected frequency:

$$\frac{(O_i - E_i)^2}{E_i}$$

This standardizes the discrepancy.

4. Sum over all categories:

$$\sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i}$$

Properties of the Chi-Square Statistic

- The statistic is always nonnegative:

$$\chi^2 \geq 0$$

- A small value of χ^2 indicates good agreement between observed and expected frequencies.
- A large value of χ^2 indicates poor agreement.
- The distribution of the statistic depends on the degrees of freedom.

Degrees of Freedom

The Chi-Square distribution used for the test depends on the degrees of freedom.

The degrees of freedom are given by:

$$df = k - p - 1$$

where:

- k = number of categories
- p = number of parameters estimated from the data

Why Degrees of Freedom are Reduced

The degrees of freedom decrease because:

1. The frequencies must sum to the total sample size:

$$\sum O_i = n$$

This imposes one restriction.

2. Additional restrictions arise when parameters are estimated from the sample.

For example:

- If no parameters are estimated:

$$df = k - 1$$

- If one parameter is estimated:

$$df = k - 2$$

- If two parameters are estimated:

$$df = k - 3$$

Conditions for Using the Test

The Chi-Square Goodness-of-Fit Test is valid under the following conditions:

1. Observations must be independent.
2. Categories must be mutually exclusive.
3. Expected frequencies should generally satisfy:

$$E_i \geq 5$$

for all categories.

4. The sample size should be sufficiently large.

Combining Categories

If some expected frequencies are too small, adjacent categories should be combined so that:

$$E_i \geq 5$$

This improves the approximation to the Chi-Square distribution.

Decision Rule

After computing the test statistic, compare it with the critical value from the Chi-Square distribution table.

Reject the null hypothesis if:

$$\chi^2_{\text{calculated}} > \chi^2_{\alpha, df}$$

Otherwise, fail to reject the null hypothesis.

Interpretation of the Decision

- If the null hypothesis is rejected, the observed data differ significantly from the hypothesized distribution.
- If the null hypothesis is not rejected, there is insufficient evidence to conclude that the hypothesized distribution is inappropriate.

P-Value Approach

Alternatively, the decision may be based on the p-value.

- Reject H_0 if:

$$p\text{-value} < \alpha$$

- Fail to reject H_0 if:

$$p\text{-value} \geq \alpha$$

Advantages of the Chi-Square Goodness-of-Fit Test

- Simple to compute and interpret.
- Applicable to many distributions.
- Does not require normality assumptions.
- Suitable for categorical data.
- Widely used in practice.

Limitations of the Chi-Square Goodness-of-Fit Test

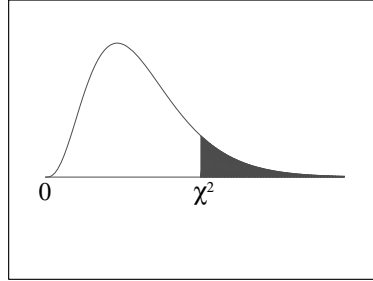
- Requires sufficiently large sample sizes.
- Sensitive to grouping of data.
- Expected frequencies should not be too small.
- Less effective for continuous data when grouping is poor.

Applications

The Chi-Square Goodness-of-Fit Test is commonly used in:

- Genetics
- Market research
- Quality control
- Reliability studies
- Biological sciences
- Social sciences
- Insurance and actuarial studies

Chi-Square Distribution Table



The shaded area is equal to α for $\chi^2 = \chi^2_{\alpha}$.

df	$\chi^2_{.995}$	$\chi^2_{.990}$	$\chi^2_{.975}$	$\chi^2_{.950}$	$\chi^2_{.900}$	$\chi^2_{.100}$	$\chi^2_{.050}$	$\chi^2_{.025}$	$\chi^2_{.010}$	$\chi^2_{.005}$
1	0.000	0.000	0.001	0.004	0.016	2.706	3.841	5.024	6.635	7.879
2	0.010	0.020	0.051	0.103	0.211	4.605	5.991	7.378	9.210	10.597
3	0.072	0.115	0.216	0.352	0.584	6.251	7.815	9.348	11.345	12.838
4	0.207	0.297	0.484	0.711	1.064	7.779	9.488	11.143	13.277	14.860
5	0.412	0.554	0.831	1.145	1.610	9.236	11.070	12.833	15.086	16.750
6	0.676	0.872	1.237	1.635	2.204	10.645	12.592	14.449	16.812	18.548
7	0.989	1.239	1.690	2.167	2.833	12.017	14.067	16.013	18.475	20.278
8	1.344	1.646	2.180	2.733	3.490	13.362	15.507	17.535	20.090	21.955
9	1.735	2.088	2.700	3.325	4.168	14.684	16.919	19.023	21.666	23.589
10	2.156	2.558	3.247	3.940	4.865	15.987	18.307	20.483	23.209	25.188
11	2.603	3.053	3.816	4.575	5.578	17.275	19.675	21.920	24.725	26.757
12	3.074	3.571	4.404	5.226	6.304	18.549	21.026	23.337	26.217	28.300
13	3.565	4.107	5.009	5.892	7.042	19.812	22.362	24.736	27.688	29.819
14	4.075	4.660	5.629	6.571	7.790	21.064	23.685	26.119	29.141	31.319
15	4.601	5.229	6.262	7.261	8.547	22.307	24.996	27.488	30.578	32.801
16	5.142	5.812	6.908	7.962	9.312	23.542	26.296	28.845	32.000	34.267
17	5.697	6.408	7.564	8.672	10.085	24.769	27.587	30.191	33.409	35.718
18	6.265	7.015	8.231	9.390	10.865	25.989	28.869	31.526	34.805	37.156
19	6.844	7.633	8.907	10.117	11.651	27.204	30.144	32.852	36.191	38.582
20	7.434	8.260	9.591	10.851	12.443	28.412	31.410	34.170	37.566	39.997
21	8.034	8.897	10.283	11.591	13.240	29.615	32.671	35.479	38.932	41.401
22	8.643	9.542	10.982	12.338	14.041	30.813	33.924	36.781	40.289	42.796
23	9.260	10.196	11.689	13.091	14.848	32.007	35.172	38.076	41.638	44.181
24	9.886	10.856	12.401	13.848	15.659	33.196	36.415	39.364	42.980	45.559
25	10.520	11.524	13.120	14.611	16.473	34.382	37.652	40.646	44.314	46.928
26	11.160	12.198	13.844	15.379	17.292	35.563	38.885	41.923	45.642	48.290
27	11.808	12.879	14.573	16.151	18.114	36.741	40.113	43.195	46.963	49.645
28	12.461	13.565	15.308	16.928	18.939	37.916	41.337	44.461	48.278	50.993
29	13.121	14.256	16.047	17.708	19.768	39.087	42.557	45.722	49.588	52.336
30	13.787	14.953	16.791	18.493	20.599	40.256	43.773	46.979	50.892	53.672
40	20.707	22.164	24.433	26.509	29.051	51.805	55.758	59.342	63.691	66.766
50	27.991	29.707	32.357	34.764	37.689	63.167	67.505	71.420	76.154	79.490
60	35.534	37.485	40.482	43.188	46.459	74.397	79.082	83.298	88.379	91.952
70	43.275	45.442	48.758	51.739	55.329	85.527	90.531	95.023	100.425	104.215
80	51.172	53.540	57.153	60.391	64.278	96.578	101.879	106.629	112.329	116.321
90	59.196	61.754	65.647	69.126	73.291	107.565	113.145	118.136	124.116	128.299
100	67.328	70.065	74.222	77.929	82.358	118.498	124.342	129.561	135.807	140.169

Example 8.3.1

A die is rolled 120 times with outcomes:

Face	1	2	3	4	5	6
Observed	14	20	18	22	24	22

Test whether the die is fair at 5% significance level.

Step 1: Hypotheses

H_0 : The die is fair

H_1 : The die is not fair

Step 2: Expected Frequencies

Since the die is fair:

$$E_i = \frac{120}{6} = 20$$

for all categories.

Step 3: Compute Test Statistic

Face	O_i	E_i	$\frac{(O_i - E_i)^2}{E_i}$
1	14	20	1.8
2	20	20	0
3	18	20	0.2
4	22	20	0.2
5	24	20	0.8
6	22	20	0.2

$$\chi^2 = 1.8 + 0 + 0.2 + 0.2 + 0.8 + 0.2 = 3.2$$

Step 4: Critical Value

Degrees of freedom:

$$df = 6 - 1 = 5$$

Critical value:

$$\chi_{0.05,5}^2 = 11.07$$

Step 5: Conclusion

Since:

$$3.2 < 11.07$$

we fail to reject H_0 .

Interpretation:

There is insufficient evidence to conclude that the die is unfair.

8.3.2 Kolmogorov–Smirnov (K–S) Test

The Kolmogorov–Smirnov (K–S) Test is a nonparametric goodness-of-fit test used to determine whether a sample of data comes from a specified probability distribution.

It is one of the most important tests for comparing probability distributions because it is based on the entire cumulative distribution function rather than grouped frequencies.

The test was developed independently by:

- Andrey Kolmogorov
- Nikolai Smirnov

The K–S test is particularly useful for continuous distributions and is widely applied in statistics, economics, engineering, reliability analysis, hydrology, and quality control.

Purpose of the Test

The K–S test is used to determine whether:

- a sample follows a specified theoretical distribution, or
- two samples come from the same distribution.

There are therefore two major forms of the test:

1. One-sample K–S test
2. Two-sample K–S test

These notes focus mainly on the one-sample K–S test.

Null and Alternative Hypotheses

Suppose $F_0(x)$ is the hypothesized cumulative distribution function.

The hypotheses are:

$$H_0 : F(x) = F_0(x)$$

against

$$H_1 : F(x) \neq F_0(x)$$

where:

- $F(x)$ = true population distribution
- $F_0(x)$ = hypothesized distribution

Basic Idea of the Test

The K-S test compares:

- the empirical distribution function obtained from the sample, and
- the theoretical cumulative distribution function under the null hypothesis.

If the sample distribution is close to the hypothesized distribution, the difference between the two functions will be small.

If the sample differs substantially from the hypothesized distribution, the difference will be large.

The test therefore measures the maximum vertical distance between the two cumulative distribution functions.

Empirical Distribution Function (EDF)

Suppose:

$$X_1, X_2, \dots, X_n$$

is a random sample.

The empirical distribution function is defined as:

$$F_n(x) = \frac{\text{Number of observations} \leq x}{n}$$

The EDF gives the proportion of sample observations less than or equal to x .

Properties of the Empirical Distribution Function

The empirical distribution function has the following properties:

1. $0 \leq F_n(x) \leq 1$
2. $F_n(x)$ is a nondecreasing function.
3. $F_n(x)$ is a step function.
4. The function jumps upward by:

$$\frac{1}{n}$$

at each observed data point.

5. As $n \rightarrow \infty$,

$$F_n(x) \rightarrow F(x)$$

for all continuity points of $F(x)$.

Construction of the Empirical Distribution Function

Suppose the observations are ordered as:

$$X_{(1)} \leq X_{(2)} \leq \cdots \leq X_{(n)}$$

Then:

$$F_n(x) = \begin{cases} 0, & x < X_{(1)} \\ \frac{1}{n}, & X_{(1)} \leq x < X_{(2)} \\ \frac{2}{n}, & X_{(2)} \leq x < X_{(3)} \\ \vdots & \\ 1, & x \geq X_{(n)} \end{cases}$$

Thus, the EDF increases in discrete jumps.

Illustration 8.3.1

Illustration of the EDF

Suppose the sample is:

$$2, 4, 5, 7$$

Then:

$$n = 4$$

The empirical distribution function becomes:

$$F_n(x) = \begin{cases} 0, & x < 2 \\ \frac{1}{4}, & 2 \leq x < 4 \\ \frac{2}{4}, & 4 \leq x < 5 \\ \frac{3}{4}, & 5 \leq x < 7 \\ 1, & x \geq 7 \end{cases}$$

K-S Test Statistic

The Kolmogorov-Smirnov test statistic is:

$$D = \sup_x |F_n(x) - F_0(x)|$$

where:

- $F_n(x)$ = empirical distribution function
- $F_0(x)$ = hypothesized cumulative distribution function
- $\sup$ denotes the supremum or maximum value

Interpretation of the Statistic

The statistic D measures the maximum absolute difference between:

- the observed cumulative proportions, and
- the expected cumulative probabilities.
- Small values of D indicate good agreement.
- Large values of D indicate poor agreement.

Thus:

- Small D supports H_0
- Large D supports H_1

Procedure for Performing the K–S Test

The general steps are:

1. State the hypotheses.
2. Arrange the observations in ascending order.
3. Compute the empirical distribution function:

$$F_n(x)$$

4. Compute the theoretical cumulative probabilities:

$$F_0(x)$$

5. Compute:

$$|F_n(x) - F_0(x)|$$

for all observations.

6. Determine the maximum absolute difference:

$$D = \max |F_n(x) - F_0(x)|$$

7. Compare the calculated value with the critical value from K–S tables.
8. Draw a conclusion.

Decision Rule

Reject the null hypothesis if:

$$D > D_\alpha$$

where:

- D_α = critical value at significance level α

Otherwise, fail to reject H_0 .

Critical Values of the Kolmogorov–Smirnov Test

After computing the test statistic:

$$D = \sup_x |F_n(x) - F_0(x)|$$

the calculated value is compared with a critical value obtained from the Kolmogorov–Smirnov critical value table.

The critical value depends on:

- the sample size n ,
- and the level of significance α .

Decision Rule

- Reject H_0 if:

$$D_{\text{calculated}} > D_{\alpha,n}$$

- Fail to reject H_0 if:

$$D_{\text{calculated}} \leq D_{\alpha,n}$$

where:

- $D_{\alpha,n}$ is the tabulated critical value corresponding to significance level α and sample size n .

Critical Value Table

For small and moderate sample sizes, exact critical values are obtained directly from the Kolmogorov–Smirnov table.

Sample Size n	Level of Significance				
	0.20	0.15	0.10	0.05	0.01
1	0.900	0.925	0.950	0.975	0.995
2	0.684	0.726	0.776	0.842	0.929
3	0.565	0.597	0.642	0.708	0.828
4	0.494	0.525	0.564	0.624	0.733
5	0.446	0.474	0.510	0.565	0.669
6	0.410	0.436	0.470	0.521	0.618
7	0.381	0.405	0.438	0.486	0.577
8	0.358	0.381	0.411	0.457	0.543
9	0.339	0.360	0.388	0.432	0.514
10	0.322	0.342	0.368	0.410	0.490
15	0.266	0.283	0.304	0.338	0.404
20	0.231	0.246	0.264	0.294	0.356
25	0.210	0.220	0.240	0.270	0.320
30	0.190	0.200	0.220	0.240	0.290
35	0.180	0.190	0.210	0.230	0.270

Approximate Critical Values for Large Samples

When the sample size exceeds 35, the critical values may be approximated using:

$$D_{\alpha} = \frac{c_{\alpha}}{\sqrt{n}}$$

where c_{α} depends on the significance level.

The commonly used constants are:

Significance Level α	c_{α}
0.20	1.07
0.15	1.14
0.10	1.22
0.05	1.36
0.01	1.63

Thus, for large samples:

$$D_{0.05} = \frac{1.36}{\sqrt{n}}$$

and

$$D_{0.01} = \frac{1.63}{\sqrt{n}}$$

Example 8.3.2

Suppose:

- sample size $n = 10$,
- significance level $\alpha = 0.05$.

From the K-S table:

$$D_{0.05,10} = 0.410$$

If the calculated test statistic is:

$$D = 0.32$$

then:

$$0.32 < 0.410$$

Hence, we fail to reject the null hypothesis.

Example 8.3.3

Large Sample Example

Suppose:

$$n = 50$$

and

$$\alpha = 0.05$$

Since $n > 35$, use the approximation:

$$\begin{aligned} D_{0.05} &= \frac{1.36}{\sqrt{50}} \\ &= \frac{1.36}{7.071} \\ &\approx 0.192 \end{aligned}$$

Therefore:

- Reject H_0 if:

$$D > 0.192$$

- Otherwise, fail to reject H_0 .

Kolmogorov-Smirnov Test Critical Values

SAMPLE SIZE (N)	LEVEL OF SIGNIFICANCE FOR D = MAXIMUM [F ₀ (X) - S _n (X)]				
	.20	.15	.10	.05	.01
1	.900	.925	.950	.975	.995
2	.684	.726	.776	.842	.929
3	.565	.597	.642	.708	.828
4	.494	.525	.564	.624	.733
5	.446	.474	.510	.565	.669
6	.410	.436	.470	.521	.618
7	.381	.405	.438	.486	.577
8	.358	.381	.411	.457	.543
9	.339	.360	.388	.432	.514
10	.322	.342	.368	.410	.490
11	.307	.326	.352	.391	.468
12	.295	.313	.338	.375	.450
13	.284	.302	.325	.361	.433
14	.274	.292	.314	.349	.418
15	.266	.283	.304	.338	.404
16	.258	.274	.295	.328	.392
17	.250	.266	.286	.318	.381
18	.244	.259	.278	.309	.371
19	.237	.252	.272	.301	.363
20	.231	.246	.264	.294	.356
25	.210	.220	.240	.270	.320
30	.190	.200	.220	.240	.290
35	.180	.190	.210	.230	.270
OVER 35	<u>1.07</u>	<u>1.14</u>	<u>1.22</u>	<u>1.36</u>	<u>1.63</u>
	$\sqrt{N}$	$\sqrt{N}$	$\sqrt{N}$	$\sqrt{N}$	$\sqrt{N}$

Example 8.3.4

Suppose a random sample of size 5 is given by:

$$2, 4, 5, 7, 9$$

We wish to test whether the data come from a uniform distribution on the interval $(0, 10)$ using the Kolmogorov–Smirnov goodness-of-fit test at the 5% level of significance.

Step 1: State the Hypotheses The null and alternative hypotheses are:

$$H_0 : X \sim U(0, 10)$$

$$H_1 : X \not\sim U(0, 10)$$

Under the null hypothesis, the cumulative distribution function of the uniform distribution on $(0, 10)$ is:

$$F_0(x) = \frac{x}{10}, \quad 0 < x < 10$$

Step 2: Arrange the Data in Ascending Order The observations are already arranged in ascending order:

$$2, 4, 5, 7, 9$$

Since:

$$n = 5$$

the empirical distribution function increases by:

$$\frac{1}{5} = 0.2$$

at each observation.

Step 3: Compute the Empirical Distribution Function For each ordered observation x_i , compute:

$$F_n(x_i) = \frac{i}{n}$$

and compare it with the theoretical cumulative probability:

$$F_0(x_i) = \frac{x_i}{10}$$

The calculations are summarized below.

i	x_i	$F_n(x_i) = \frac{i}{5}$	$F_0(x_i) = \frac{x_i}{10}$	$ F_n(x_i) - F_0(x_i) $
1	2	0.2	0.2	0.0
2	4	0.4	0.4	0.0
3	5	0.6	0.5	0.1
4	7	0.8	0.7	0.1
5	9	1.0	0.9	0.1

Step 4: Compute the K–S Test Statistic The Kolmogorov–Smirnov test statistic is:

$$D = \sup_x |F_n(x) - F_0(x)|$$

From the table above, the maximum absolute difference is:

$$D = 0.1$$

Step 5: Obtain the Critical Value Using the Kolmogorov–Smirnov critical value table for:

$$n = 5$$

and significance level:

$$\alpha = 0.05$$

the critical value is:

$$D_{0.05,5} = 0.565$$

Step 6: Make the Decision Compare the calculated statistic with the tabulated critical value:

$$D_{\text{calculated}} = 0.1$$

$$D_{\text{critical}} = 0.565$$

Since:

$$0.1 < 0.565$$

we fail to reject the null hypothesis.

Conclusion There is insufficient evidence at the 5% significance level to conclude that the data do not follow a uniform distribution on $(0, 10)$. Hence, the sample is consistent with the distribution:

$$X \sim U(0, 10)$$

8.4 Order Statistics

Order statistics are among the most important concepts in probability theory and non-parametric statistics.

They arise whenever observations from a sample are arranged according to magnitude. Many statistical procedures, especially nonparametric methods, are based on ordered observations rather than the raw data themselves.

Order statistics play a fundamental role in:

- nonparametric inference,
- reliability theory,
- survival analysis,
- extreme value theory,
- quality control,
- actuarial science,
- robust statistics.

They are especially useful in studying extreme observations such as the smallest or largest values in a sample.

Definition

Suppose:

$$X_1, X_2, \dots, X_n$$

is a random sample drawn from a population with probability density function $f(x)$ and cumulative distribution function $F(x)$.

The observations may initially appear in any order:

$$X_1, X_2, X_3, \dots, X_n$$

If the observations are rearranged in ascending order of magnitude, we obtain:

$$X_{(1)} \leq X_{(2)} \leq \dots \leq X_{(n)}$$

These ordered observations are called **order statistics**.

The notation:

$$X_{(r)}$$

denotes the r -th smallest observation in the sample.

Important Note The parentheses in:

$$X_{(r)}$$

are used specifically to distinguish order statistics from ordinary indexed observations such as X_r .

- X_r denotes the r -th observation in the original sample.
- $X_{(r)}$ denotes the r -th smallest observation after ordering.

Illustration

Suppose the sample is:

$$7, 2, 10, 5, 1$$

The observations in ascending order become:

$$1, 2, 5, 7, 10$$

Hence:

$$X_{(1)} = 1$$

$$X_{(2)} = 2$$

$$X_{(3)} = 5$$

$$X_{(4)} = 7$$

$$X_{(5)} = 10$$

Terminology

Some order statistics have special names and interpretations.

Statistic	Meaning
$X_{(1)}$	Minimum observation
$X_{(n)}$	Maximum observation
$X_{(2)}$	Second smallest observation
$X_{(n-1)}$	Second largest observation
$X_{((n+1)/2)}$	Median (for odd n)

Minimum The first order statistic:

$$X_{(1)}$$

represents the smallest value in the sample.

Maximum The last order statistic:

$$X_{(n)}$$

represents the largest value in the sample.

Median When n is odd, the middle order statistic:

$$X_{(\frac{n+1}{2})}$$

is the sample median.

For even n , the median is usually defined as the average of the two middle observations.

8.4.1 Distribution of Order Statistics

To use order statistics effectively, it is important to understand their probability distributions.

Suppose:

$$X_1, X_2, \dots, X_n$$

is a random sample from a continuous distribution with:

$$f(x) = \text{pdf}$$

and

$$F(x) = \text{cdf}$$

The distribution of:

$$X_{(r)}$$

can be derived using probability arguments.

Idea Behind the Distribution

To determine the probability density function of the r -th order statistic, consider the following reasoning.

For:

$$X_{(r)} = x$$

to occur:

- exactly $r - 1$ observations must be less than x ,
- one observation must lie near x ,
- and $n - r$ observations must exceed x .

These probabilities combine to produce the density function of the order statistic.

PDF of the r -th Order Statistic

The probability density function of the r -th order statistic is:

$$f_{X_{(r)}}(x) = \frac{n!}{(r-1)!(n-r)!} [F(x)]^{r-1} [1-F(x)]^{n-r} f(x)$$

where:

- n = sample size,
- r = order of the statistic,
- $f(x)$ = parent probability density function,
- $F(x)$ = parent cumulative distribution function.

Explanation of the Formula

The formula contains several important components.

1. Combinatorial Term

$$\frac{n!}{(r-1)!(n-r)!}$$

counts the number of ways of choosing:

- $r - 1$ observations below x ,
- and $n - r$ observations above x .

2. Probability That $r - 1$ Observations Are Below x

$$[F(x)]^{r-1}$$

represents the probability that $r - 1$ observations fall below x .

3. Probability That $n - r$ Observations Are Above x

$$[1 - F(x)]^{n-r}$$

represents the probability that the remaining observations exceed x .

4. Density at x

$$f(x)$$

accounts for the probability of an observation occurring near x .

Special Cases of Order Statistics

Some order statistics are especially important in practice.

- Minimum
- Maximum
- Median

Distribution of the Minimum

The minimum observation is:

$$X_{(1)}$$

Setting:

$$r = 1$$

in the general formula gives:

$$f_{X_{(1)}}(x) = n[1 - F(x)]^{n-1}f(x)$$

Interpretation For the minimum to occur near x :

- one observation must occur near x ,
- all remaining observations must exceed x .

The term:

$$[1 - F(x)]^{n-1}$$

represents the probability that all other observations are larger than x .

CDF of the Minimum

The cumulative distribution function of the minimum is:

$$F_{X_{(1)}}(x) = 1 - [1 - F(x)]^n$$

This follows because:

$$P(X_{(1)} > x) = [1 - F(x)]^n$$

Interpretation The probability that the minimum exceeds x means that all observations exceed x .

Distribution of the Maximum

The maximum observation is:

$$X_{(n)}$$

Setting:

$$r = n$$

in the general formula gives:

$$f_{X_{(n)}}(x) = n[F(x)]^{n-1}f(x)$$

Interpretation For the maximum to occur near x :

- one observation must occur near x ,
- all remaining observations must be less than x .

The term:

$$[F(x)]^{n-1}$$

represents the probability that all remaining observations are below x .

CDF of the Maximum

The cumulative distribution function of the maximum is:

$$F_{X_{(n)}}(x) = [F(x)]^n$$

because:

$$P(X_{(n)} \leq x) = P(X_1 \leq x, \dots, X_n \leq x)$$

and independence gives:

$$[F(x)]^n$$

Behavior of Extreme Order Statistics

As the sample size increases:

- the minimum tends to move toward the lower tail,
- the maximum tends to move toward the upper tail.

Thus, larger samples are more likely to contain extreme observations.

Applications of Minimum and Maximum Order Statistics

The minimum and maximum are used extensively in:

- reliability analysis,
- life testing,
- environmental extremes,
- flood analysis,
- financial risk management,
- quality control.

Advantages of Order Statistics

Order statistics possess several important advantages.

1. They are distribution-free in many applications.
2. They are robust against outliers.
3. They form the basis of nonparametric methods.
4. They are useful for studying extremes.

Summary

Order statistics are ordered sample observations obtained by arranging data in ascending order.

The ordered sample is:

$$X_{(1)} \leq X_{(2)} \leq \cdots \leq X_{(n)}$$

The general density of the r -th order statistic is:

$$f_{X_{(r)}}(x) = \frac{n!}{(r-1)!(n-r)!} [F(x)]^{r-1} [1-F(x)]^{n-r} f(x)$$

Important special cases include:

- minimum:

$$f_{X_{(1)}}(x) = n[1-F(x)]^{n-1}f(x)$$

- maximum:

$$f_{X_{(n)}}(x) = n[F(x)]^{n-1}f(x)$$

Order statistics are fundamental tools in probability theory, statistical inference, and nonparametric statistics.

Example 8.4.1

Suppose:

$$X_1, X_2, X_3 \sim U(0, 1)$$

are independent and identically distributed random variables from the uniform distribution on the interval $(0, 1)$.

We wish to find the distribution of the maximum order statistic:

$$X_{(3)} = \max(X_1, X_2, X_3)$$

That is, we want to determine:

- the cumulative distribution function (CDF),
- and the probability density function (PDF)

of the maximum.

Step 1: Identify the Parent Distribution

Since:

$$X_i \sim U(0, 1)$$

the probability density function of each observation is:

$$f(x) = \begin{cases} 1, & 0 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

The cumulative distribution function is:

$$F(x) = P(X_i \leq x)$$

For the uniform distribution on $(0, 1)$:

$$F(x) = \begin{cases} 0, & x \leq 0 \\ x, & 0 < x < 1 \\ 1, & x \geq 1 \end{cases}$$

Step 2: Define the Maximum Order Statistic

The maximum observation is:

$$X_{(3)} = \max(X_1, X_2, X_3)$$

We seek the distribution of:

$$X_{(3)}$$

To obtain the distribution, we first derive the cumulative distribution function.

Step 3: Find the CDF of the Maximum

By definition:

$$F_{X_{(3)}}(x) = P(X_{(3)} \leq x)$$

The event:

$$X_{(3)} \leq x$$

means that the largest observation is less than or equal to x .
This can only happen if all observations are less than or equal to x .
Thus:

$$P(X_{(3)} \leq x) = P(X_1 \leq x, X_2 \leq x, X_3 \leq x)$$

Step 4: Use Independence

Since the random variables are independent:

$$P(X_1 \leq x, X_2 \leq x, X_3 \leq x) = P(X_1 \leq x)P(X_2 \leq x)P(X_3 \leq x)$$

Because all variables have the same distribution:

$$= [F(x)]^3$$

For the uniform distribution:

$$F(x) = x, \quad 0 < x < 1$$

Hence:

$$F_{X_{(3)}}(x) = x^3, \quad 0 < x < 1$$

Therefore, the cumulative distribution function of the maximum is:

$$F_{X_{(3)}}(x) = \begin{cases} 0, & x \leq 0 \\ x^3, & 0 < x < 1 \\ 1, & x \geq 1 \end{cases}$$

Step 5: Differentiate to Obtain the PDF

The probability density function is obtained by differentiating the cumulative distribution function.

Thus:

$$f_{X_{(3)}}(x) = \frac{d}{dx} F_{X_{(3)}}(x)$$

For:

$$0 < x < 1$$

we have:

$$F_{X_{(3)}}(x) = x^3$$

Differentiating:

$$f_{X_{(3)}}(x) = \frac{d}{dx}(x^3)$$

$$= 3x^2$$

Therefore, the probability density function of the maximum is:

$$f_{X_{(3)}}(x) = \begin{cases} 3x^2, & 0 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

Step 6: Verify That the PDF is Valid

To verify that this is a valid density function, check that:

$$\int_{-\infty}^{\infty} f_{X_{(3)}}(x) dx = 1$$

Since the density is nonzero only on $(0, 1)$:

$$\begin{aligned} \int_0^1 3x^2 dx &= 3 \int_0^1 x^2 dx \\ &= 3 \left[\frac{x^3}{3} \right]_0^1 \\ &= [x^3]_0^1 \\ &= 1 - 0 \\ &= 1 \end{aligned}$$

Hence, the density is valid.

Interpretation

The density:

$$f_{X_{(3)}}(x) = 3x^2$$

places more probability near $x = 1$.

This is reasonable because the maximum of several observations from $U(0, 1)$ is likely to be relatively large.

Alternative Method Using the General Formula

The same result can be obtained using the general formula for the maximum order statistic:

$$f_{X_{(n)}}(x) = n[F(x)]^{n-1}f(x)$$

Here:

$$n = 3$$

$$F(x) = x$$

$$f(x) = 1$$

Substituting:

$$f_{X_{(3)}}(x) = 3(x)^2(1)$$

$$= 3x^2$$

which agrees with the previous result.

Final Answer

The cumulative distribution function of the maximum is:

$$F_{X_{(3)}}(x) = \begin{cases} 0, & x \leq 0 \\ x^3, & 0 < x < 1 \\ 1, & x \geq 1 \end{cases}$$

The probability density function is:

$$f_{X_{(3)}}(x) = \begin{cases} 3x^2, & 0 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

Example 8.4.2

[Distribution of the Minimum Order Statistic]

Suppose:

$$X_1, X_2, X_3, X_4 \sim U(0, 2)$$

are independent and identically distributed random variables.

Find the distribution of the minimum order statistic:

$$X_{(1)} = \min(X_1, X_2, X_3, X_4)$$

Step 1: Identify the Parent Distribution

Since:

$$X_i \sim U(0, 2)$$

the probability density function is:

$$f(x) = \begin{cases} \frac{1}{2}, & 0 < x < 2 \\ 0, & \text{otherwise} \end{cases}$$

The cumulative distribution function is:

$$F(x) = P(X_i \leq x) = \begin{cases} 0, & x \leq 0 \\ \frac{x}{2}, & 0 < x < 2 \\ 1, & x \geq 2 \end{cases}$$

Step 2: Define the Minimum Order Statistic

The minimum observation is:

$$X_{(1)} = \min(X_1, X_2, X_3, X_4)$$

We first derive the cumulative distribution function of the minimum.

Step 3: Derive the CDF of the Minimum

By definition:

$$F_{X_{(1)}}(x) = P(X_{(1)} \leq x)$$

It is usually easier to first compute:

$$P(X_{(1)} > x)$$

The event:

$$X_{(1)} > x$$

means that all observations exceed x .

Thus:

$$P(X_{(1)} > x) = P(X_1 > x, X_2 > x, X_3 > x, X_4 > x)$$

Since the variables are independent:

$$= P(X_1 > x)P(X_2 > x)P(X_3 > x)P(X_4 > x)$$

$$= [1 - F(x)]^4$$

For the uniform distribution:

$$F(x) = \frac{x}{2}$$

Hence:

$$P(X_{(1)} > x) = \left(1 - \frac{x}{2}\right)^4$$

for:

$$0 < x < 2$$

Therefore:

$$F_{X_{(1)}}(x) = 1 - \left(1 - \frac{x}{2}\right)^4$$

Thus, the cumulative distribution function is:

$$F_{X_{(1)}}(x) = \begin{cases} 0, & x \leq 0 \\ 1 - \left(1 - \frac{x}{2}\right)^4, & 0 < x < 2 \\ 1, & x \geq 2 \end{cases}$$

Step 4: Differentiate to Obtain the PDF

Differentiate the CDF:

$$f_{X_{(1)}}(x) = \frac{d}{dx} F_{X_{(1)}}(x)$$

Thus:

$$f_{X_{(1)}}(x) = \frac{d}{dx} \left[1 - \left(1 - \frac{x}{2}\right)^4 \right]$$

Applying the chain rule:

$$\begin{aligned} &= -4 \left(1 - \frac{x}{2}\right)^3 \left(-\frac{1}{2}\right) \\ &= 2 \left(1 - \frac{x}{2}\right)^3 \end{aligned}$$

for:

$$0 < x < 2$$

Hence:

$$f_{X_{(1)}}(x) = \begin{cases} 2 \left(1 - \frac{x}{2}\right)^3, & 0 < x < 2 \\ 0, & \text{otherwise} \end{cases}$$

Step 5: Interpretation

The density is largest near:

$$x = 0$$

This is expected because the minimum of several observations is likely to be small.

Final Answer

The cumulative distribution function is:

$$F_{X_{(1)}}(x) = \begin{cases} 0, & x \leq 0 \\ 1 - \left(1 - \frac{x}{2}\right)^4, & 0 < x < 2 \\ 1, & x \geq 2 \end{cases}$$

The probability density function is:

$$f_{X_{(1)}}(x) = \begin{cases} 2 \left(1 - \frac{x}{2}\right)^3, & 0 < x < 2 \\ 0, & \text{otherwise} \end{cases}$$

Example 8.4.3

[Distribution of the Median Order Statistic]

Suppose:

$$X_1, X_2, X_3 \sim U(0, 1)$$

are independent and identically distributed random variables.

Find the distribution of the median order statistic:

$$X_{(2)}$$

Step 1: Identify the Parent Distribution

Since:

$$X_i \sim U(0, 1)$$

the probability density function is:

$$f(x) = \begin{cases} 1, & 0 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

The cumulative distribution function is:

$$F(x) = \begin{cases} 0, & x \leq 0 \\ x, & 0 < x < 1 \\ 1, & x \geq 1 \end{cases}$$

Step 2: Use the General Formula

The density of the r -th order statistic is:

$$f_{X_{(r)}}(x) = \frac{n!}{(r-1)!(n-r)!} [F(x)]^{r-1} [1 - F(x)]^{n-r} f(x)$$

For the median:

$$r = 2$$

and:

$$n = 3$$

Substituting into the formula:

$$f_{X_{(2)}}(x) = \frac{3!}{(2-1)!(3-2)!} [F(x)]^{2-1} [1-F(x)]^{3-2} f(x)$$

Step 3: Simplify the Formula

Compute the factorial term:

$$\frac{3!}{1!1!} = \frac{6}{1} = 6$$

Thus:

$$f_{X_{(2)}}(x) = 6[F(x)][1-F(x)]f(x)$$

Since:

$$F(x) = x$$

and

$$f(x) = 1$$

for:

$$0 < x < 1$$

we obtain:

$$f_{X_{(2)}}(x) = 6x(1-x)$$

for:

$$0 < x < 1$$

Hence:

$$f_{X_{(2)}}(x) = \begin{cases} 6x(1-x), & 0 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

Step 4: Verify That the Density is Valid

Check that:

$$\int_0^1 6x(1-x) dx = 1$$

Expanding:

$$\begin{aligned} &= 6 \int_0^1 (x - x^2) dx \\ &= 6 \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_0^1 \end{aligned}$$

$$= 6 \left(\frac{1}{2} - \frac{1}{3} \right)$$

$$= 6 \left(\frac{1}{6} \right)$$

$$= 1$$

Thus, the density is valid.

Step 5: Interpretation

The density:

$$f_{X_{(2)}}(x) = 6x(1 - x)$$

is symmetric about:

$$x = \frac{1}{2}$$

This is reasonable because the median of three uniform observations is most likely to lie near the center of the interval.

Final Answer

The probability density function of the median order statistic is:

$$f_{X_{(2)}}(x) = \begin{cases} 6x(1 - x), & 0 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

8.5 Measures of Association

In statistics, it is often important to determine whether two variables are related and to measure the strength of that relationship.

Measures of association are statistical tools used to quantify the degree of relationship between variables.

They help answer questions such as:

- Do two variables move together?
- Does an increase in one variable correspond to an increase or decrease in another?
- How strong is the relationship?
- Is the relationship positive or negative?

Types of Association

The relationship between variables may take different forms.

Positive Association A positive association occurs when large values of one variable tend to be associated with large values of another variable.

For example:

- income and expenditure,
- height and weight,
- study time and examination scores.

Negative Association A negative association occurs when large values of one variable tend to be associated with small values of another variable.

Examples include:

- price and demand,
- speed and travel time,
- stress and productivity.

No Association No association exists when changes in one variable are unrelated to changes in another variable.

Linear and Monotonic Relationships

Relationships between variables may be:

- linear,
- nonlinear,
- monotonic,
- non-monotonic.

Linear Relationship A linear relationship occurs when the variables change approximately along a straight line.

Such relationships are commonly measured using Pearson's correlation coefficient.

Monotonic Relationship A monotonic relationship occurs when one variable generally increases or decreases as the other increases, though not necessarily at a constant rate.

For example:

- as income increases, expenditure may generally increase,
- but not necessarily linearly.

Nonparametric association measures are especially useful for monotonic relationships.

Nonparametric Measures of Association

Nonparametric association measures are methods that do not require strong assumptions about the underlying distributions of the variables.

They are particularly useful when:

- data are ordinal,
- the normality assumption fails,
- outliers are present,
- relationships are monotonic but not linear,
- sample sizes are small.

Unlike Pearson correlation, nonparametric methods rely mainly on ranks rather than the original numerical values.

Advantages of Nonparametric Association Measures

Nonparametric measures possess several advantages.

1. They are robust against outliers.
2. They can be used for ordinal data.
3. They do not require normality assumptions.
4. They can detect monotonic relationships.
5. They are easy to interpret.

8.5.1 Spearman Rank Correlation

Spearman's rank correlation coefficient is one of the most widely used nonparametric measures of association.

It was developed by Charles Spearman and measures the strength and direction of association between two ranked variables.

The statistic is usually denoted by:

$$r_s$$

or sometimes:

$$\rho_s$$

Basic Idea of Spearman Correlation

Instead of using the original numerical values, Spearman correlation converts the observations into ranks.

The relationship between the ranks is then analyzed.

If two variables have very similar rankings, the association is strong and positive.

If one variable increases while the other decreases, the association is negative.

When Spearman Correlation is Used

Spearman rank correlation is appropriate when:

- data are ordinal,
- variables are measured as ranks,
- normality assumptions are violated,
- the relationship is monotonic,
- outliers affect Pearson correlation.

Procedure for Computing Spearman Correlation

The computation involves the following steps.

1. Rank the observations for each variable.
2. Compute the difference between paired ranks.
3. Square the rank differences.

4. Sum the squared differences.
5. Substitute into the Spearman formula.

Notation

Suppose:

- R_i = rank of observation i for variable X ,
- S_i = rank of observation i for variable Y .

Define:

$$d_i = R_i - S_i$$

Then:

$$d_i^2 = (R_i - S_i)^2$$

Formula

Spearman's rank correlation coefficient is:

$$r_s = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}$$

where:

- d_i = difference between paired ranks,
- n = number of paired observations.

Interpretation of Intermediate Values

Values of r_s between -1 and 1 indicate varying degrees of association.

Value of r_s	Interpretation
0.00 – 0.19	Very weak association
0.20 – 0.39	Weak association
0.40 – 0.59	Moderate association
0.60 – 0.79	Strong association
0.80 – 1.00	Very strong association

The same interpretation applies to negative values.

9 Worked Example: Spearman Correlation

Student	Math Rank	Statistics Rank
A	1	2
B	2	1
C	3	4
D	4	3
E	5	5

Step 1: Compute Rank Differences

Student	d_i	d_i^2
A	-1	1
B	1	1
C	-1	1
D	1	1
E	0	0

$$\sum d_i^2 = 4$$

Step 2: Compute Correlation

$$\begin{aligned} r_s &= 1 - \frac{6(4)}{5(25 - 1)} \\ &= 1 - \frac{24}{120} \\ &= 0.8 \end{aligned}$$

Interpretation:

There is a strong positive association between the two rankings.

9.0.1 Kendall's Tau

Kendall's tau is an important nonparametric measure of association used to determine the strength and direction of relationship between two variables.

It was developed by Maurice Kendall and is particularly useful when data are measured on an ordinal scale or when the assumptions required for Pearson correlation are not satisfied.

Like Spearman's rank correlation coefficient, Kendall's tau is based on ranks rather than the original numerical values. However, instead of using rank differences, Kendall's tau measures association by comparing pairs of observations.

The coefficient is commonly denoted by:

$$\tau$$

or sometimes:

$$\tau_K$$

Purpose of Kendall's Tau

Kendall's tau is used to:

- measure the strength of association between two variables,
- determine whether rankings are similar,
- detect monotonic relationships,
- analyze ordinal data,
- assess agreement between judges or raters.

Situations Where Kendall's Tau is Appropriate

Kendall's tau is especially useful when:

- data are ordinal,
- sample sizes are small,
- the relationship is monotonic,
- there are tied observations,
- assumptions of normality are violated.

Basic Idea Behind Kendall's Tau

The basic idea behind Kendall's tau is to examine whether pairs of observations are ordered in the same direction.

Suppose two variables are:

$$X \quad \text{and} \quad Y$$

If higher values of X tend to correspond to higher values of Y , then the variables exhibit positive association.

If higher values of X correspond to lower values of Y , then the variables exhibit negative association.

Kendall's tau measures this association by comparing all possible pairs of observations.

Concordant and Discordant Pairs

The concepts of concordant and discordant pairs are central to Kendall's tau.

Consider two observations:

$$(X_i, Y_i) \quad \text{and} \quad (X_j, Y_j)$$

Concordant Pairs A pair is called **concordant** if the rankings of both variables agree.

This occurs when:

$$(X_i - X_j)(Y_i - Y_j) > 0$$

That is:

- if $X_i > X_j$ and $Y_i > Y_j$,
- or if $X_i < X_j$ and $Y_i < Y_j$.

Thus, the observations move in the same direction.

Example of Concordant Pair Suppose:

$$(X_i, Y_i) = (5, 8)$$

and

$$(X_j, Y_j) = (3, 4)$$

Since:

$$5 > 3$$

and

$$8 > 4$$

the pair is concordant.

Discordant Pairs A pair is called **discordant** if the rankings disagree.

This occurs when:

$$(X_i - X_j)(Y_i - Y_j) < 0$$

That is:

- if $X_i > X_j$ but $Y_i < Y_j$,
- or if $X_i < X_j$ but $Y_i > Y_j$.

Thus, the observations move in opposite directions.

Example of Discordant Pair Suppose:

$$(X_i, Y_i) = (6, 2)$$

and

$$(X_j, Y_j) = (4, 7)$$

Since:

$$6 > 4$$

but

$$2 < 7$$

the pair is discordant.

Total Number of Pairs

For a sample of size:

$$n$$

the total number of distinct pairs is:

$$\frac{n(n-1)}{2}$$

This represents the denominator in Kendall's tau formula.

Formula

Kendall's tau coefficient is:

$$\tau = \frac{C - D}{\frac{1}{2}n(n-1)}$$

where:

- C = number of concordant pairs,
- D = number of discordant pairs,
- $\frac{n(n-1)}{2}$ = total number of pairs.

Explanation of the Formula

The quantity:

$$C - D$$

measures the difference between agreement and disagreement among pairs.

- If concordant pairs dominate, the coefficient is positive.
- If discordant pairs dominate, the coefficient is negative.
- If concordant and discordant pairs are approximately equal, the coefficient is near zero.

Dividing by the total number of pairs standardizes the coefficient so that it lies between:

$$-1 \quad \text{and} \quad 1$$

Properties of Kendall's Tau

Kendall's tau satisfies:

$$-1 \leq \tau \leq 1$$

Perfect Positive Association If:

$$\tau = 1$$

all pairs are concordant.

This means the rankings are in perfect agreement.

Perfect Negative Association If:

$$\tau = -1$$

all pairs are discordant.

This means the rankings are completely reversed.

No Association If:

$$\tau = 0$$

the numbers of concordant and discordant pairs are approximately equal.

This indicates no monotonic association.

Interpretation of Kendall's Tau

Kendall's tau has an intuitive probabilistic interpretation.

It may be interpreted as:

The difference between the probability of concordance and the probability of discordance.

That is:

$$\tau = P(\text{concordance}) - P(\text{discordance})$$

This interpretation makes Kendall's tau especially meaningful in practical applications.

Procedure for Computing Kendall's Tau

The general procedure is:

1. Arrange the paired observations.
2. Form all possible pairs.
3. Determine whether each pair is concordant or discordant.
4. Count:

$$C \quad \text{and} \quad D$$

5. Substitute into the formula:

$$\tau = \frac{C - D}{\frac{1}{2}n(n-1)}$$

Handling Ties

A tie occurs when two observations have equal values.

For example:

$$X_i = X_j$$

or

$$Y_i = Y_j$$

In the presence of ties, modified versions such as:

$$\tau_b$$

or

$$\tau_c$$

are often used.

These versions adjust for tied observations.

Advantages of Kendall's Tau

Kendall's tau possesses several important advantages.

1. It does not require normality assumptions.
2. It is suitable for ordinal data.
3. It has a clear probabilistic interpretation.
4. It is robust against outliers.
5. It performs well for small samples.
6. It handles ties better than Spearman correlation.

Limitations of Kendall's Tau

Despite its usefulness, Kendall's tau also has some limitations.

1. Computation becomes intensive for large samples.
2. It may be less familiar than Pearson correlation.
3. It only measures monotonic relationships.
4. It uses ranks rather than exact numerical magnitudes.

Comparison Between Spearman and Kendall

Both Spearman and Kendall are nonparametric measures of association, but they differ in several ways.

Feature	Spearman	Kendall
Based on	Rank differences	Pair comparisons
Interpretation	Correlation-like	Probability-based
Sensitivity to ties	Moderate	Lower
Computation	Easier	More intensive
Robustness	Moderate	Higher
Small sample performance	Good	Excellent

Spearman versus Kendall

In practice:

- Spearman correlation is easier to compute.
- Kendall's tau provides better theoretical interpretation.
- Kendall's tau is often preferred for small samples and tied observations.
- Spearman correlation is more commonly used in introductory applications.

Example 9.0.1

[Kendall's Tau Rank Correlation]

Suppose five students are ranked in Mathematics and Statistics as follows:

Student	Mathematics Rank (X)	Statistics Rank (Y)
A	1	2
B	2	1
C	3	4
D	4	3
E	5	5

Compute Kendall's tau coefficient to determine the association between the two rankings.

Step 1: State the Objective

We wish to measure the strength and direction of association between the Mathematics and Statistics rankings using Kendall's tau.

The formula is:

$$\tau = \frac{C - D}{\frac{1}{2}n(n - 1)}$$

where:

- C = number of concordant pairs,
- D = number of discordant pairs,
- n = number of observations.

Step 2: Determine the Number of Pairs

Since:

$$n = 5$$

the total number of possible pairs is:

$$\begin{aligned}\frac{1}{2}n(n - 1) &= \frac{1}{2}(5)(4) \\ &= 10\end{aligned}$$

Thus, there are 10 pairs to examine.

Step 3: List All Possible Pairs

The possible pairs are:

$$(A, B), (A, C), (A, D), (A, E),$$

$$(B, C), (B, D), (B, E),$$

$$(C, D), (C, E),$$

$$(D, E)$$

We now determine whether each pair is concordant or discordant.

Step 4: Determine Concordant and Discordant Pairs

Pair (A, B)

$$X : 1 < 2$$

but

$$Y : 2 > 1$$

The rankings disagree.

Therefore, the pair is discordant.

$$D = 1$$

Pair (A, C)

$$X : 1 < 3$$

and

$$Y : 2 < 4$$

The rankings agree.
Thus, the pair is concordant.

$$C = 1$$

Pair (A, D)

$$X : 1 < 4$$

and

$$Y : 2 < 3$$

The pair is concordant.

$$C = 2$$

Pair (A, E)

$$X : 1 < 5$$

and

$$Y : 2 < 5$$

The pair is concordant.

$$C = 3$$

Pair (B, C)

$$X : 2 < 3$$

and

$$Y : 1 < 4$$

The pair is concordant.

$$C = 4$$

Pair (B, D)

$$X : 2 < 4$$

and

$$Y : 1 < 3$$

The pair is concordant.

$$C = 5$$

Pair (B, E)

$$X : 2 < 5$$

and

$$Y : 1 < 5$$

The pair is concordant.

$$C = 6$$

Pair (C, D)

$$X : 3 < 4$$

but

$$Y : 4 > 3$$

The pair is discordant.

$$D = 2$$

Pair (C, E)

$$X : 3 < 5$$

and

$$Y : 4 < 5$$

The pair is concordant.

$$C = 7$$

Pair (D, E)

$$X : 4 < 5$$

and

$$Y : 3 < 5$$

The pair is concordant.

$$C = 8$$

Step 5: Summarize the Counts

The total numbers are:

$$C = 8$$

$$D = 2$$

The total number of pairs is:

$$10$$

Step 6: Compute Kendall's Tau

Substitute into the formula:

$$\begin{aligned}\tau &= \frac{C - D}{\frac{1}{2}n(n - 1)} \\ &= \frac{8 - 2}{10} \\ &= \frac{6}{10} \\ &= 0.6\end{aligned}$$

Step 7: Interpretation

The Kendall's tau coefficient is:

$$\tau = 0.6$$

This indicates:

- a positive association between the two rankings,
- and a moderately strong relationship.

Thus, students who perform well in Mathematics also tend to perform well in Statistics.

Conclusion

There is a moderately strong positive association between the Mathematics and Statistics rankings.

Hence, the two sets of rankings are fairly consistent.